Earners AI and ML - by Aditya Jain Sir - Lecture 07

Recap = for loop: Quiz

```
Q1. How many times will the following loop execute?

for i in range(5):
    print("Hello")


A. 4

B. 5

C. 6

D. Infinite

Ans: B
```

```
for i in range(5):
    print("Hello")
```

```
Hello
Hello
Hello
Hello
Hello
```

```
Q2. What will be the output?

for i in range(4):
    print(i)

A.

1
2
3
4

B.

0
1
2
3

C.

0
1
2
3
4

D. Error

Ans: B
```

```
Q3. What numbers are generated by range(3, 8)?

A.

3 4 5 6 7

B.

3 4 5 6 7 8
```

C.

4 5 6 7 8

D.

3 4 5 6

Ans: A

What will be printed?

```
for i in range(2, 11, 2):
    print(i)
```

A.

2
4
6
8
10

B.

2
4
6
8
10
12

C.

1
3
5
7
9

D. Error

Ans: A

```
What is the output?
for i in range(10, 0, -2):
    print(i)

A.

10
8
6
4
2

B.

10
8
6
4
2
0

C.

10
9
8
7
6

D. Error

Ans: A
```

```
What does range(5) generate?

A.

1 2 3 4 5

B.

0 1 2 3 4

C.

0 1 2 3 4 5

D.

5

Ans: B
```

```
Q7. Which keyword is used to create a for loop?

A. repeat

B. while

C. for

D. loop

Ans: C
```

```
Q8. What is the output?
for i in range(1, 6):
    print(i * i)

A.

1
4
9
16
```

```
25

B.

2
4
6
8
10

C.

0
1
4
9
16

D. Error

Ans : A
```

```
Q9. What will this print?
for ch in "Python":
    print(ch)

A.

Python

B.

Error

C.

P y t h o n

D.

P
y
t
h
o
n
```

```
for ch in "Python":
    print(ch)
```

```
P
y
t
h
o
n
```

```
for ch in "Python":
    print(ch, end  = " ")
```

```
P y t h o n
```

```
for ch in "Python":
    print(ch, end  = "")
```

```
Python
```

```
Which of the following is the correct syntax?

A.

for i range(5):
```

B.

for(i in range(5)):

C.

for i in range(5):

D.

for range(5):

Ans: C

What will be the output?

```
for i in range(5, 1, -1):
    print(i)
```

A.

5
4
3
2

B.

5
4
3
2
1

C.

1
2
3
4
5

D. Error

Ans: A

```
for i in range(1, 10, 3):
    print(i)
```

A.

1
4
7

B.

1
3
6
9

C.

1
4
7
10

D. 7, 10

Ans: A

```
How many times will the following loop execute?
for i in range(1, 11, 2):
    print(i)

A. 4

B. 5

C. 6

D. 10

Ans : B
```

```
What is the output?
for i in range(3, 3):
    print(i)

A.

3

B.

0

C. Nothing

D. Error

Ans : C
```

```
for i in range(3, 3):
    print(i)
```

```
Which statement about range(start, stop) is TRUE?

A. Both start and stop are included.

B. Only stop is included.

C. start is included, stop is excluded.

D. Both are excluded.

Ans: C
```

```
What is the output?
for i in range(4):
    print(i + 10)

A.

10
11
12
13

B.

11
12
13
14

C.

10
20
30
40

D. Error
```

```
Ans: A
```

```
Predict the output.
for i in range(2):
    print("Python")
    print("AI")

A.

Python
AI

B.

Python
AI
Python
AI

C.

Python
Python
AI
AI

D. Infinite Loop

Ans: B
```

```
for i in range(2):
    print("Python")
    print("AI")
```

```
Python
AI
Python
AI
```

```
for i in range(2):
    print("Python")
print("AI")
```

```
Python
Python
AI
```

```
name = "Data"

for ch in name:
    if ch == "a":
        print(ch)

A.

a

B.

a
a

C.

Data

D. Error
```

```
name = "Data"

for ch in name:
```

```
    if ch == "a":
        print(ch)
```

```
a
a
```

```
name = "Data"

for ch in name:
    if ch != "a":
        print(ch)
```

```
D
t
```

While loop

```
# 1. initialization
# 2. condition
# 3. updation
```

```
#Print all the int from 1 to 5.

for i in range(1,6):
  print(i)
```

```
1
2
3
4
5
```

```
i = 1

while i<6:
  print(i)
  i = i+1
```

```
1
2
3
4
5
```

```
#Write a while loop to print Hi 7 times on the screen.

i = 1
while i<=7:
  print(i, "Hi")
  i = i+1
```

```
1 Hi
2 Hi
3 Hi
4 Hi
5 Hi
6 Hi
7 Hi
```

```
i = 1
while i<=7:
  print(i, "Hi")
  i += 1 # i = i+1
```

```
1 Hi
2 Hi
3 Hi
4 Hi
5 Hi
6 Hi
7 Hi
```

```
i = 7
while i>=1:
```

```
  print(i, "Hi")
  i = i-1
```

```
7 Hi
6 Hi
5 Hi
4 Hi
3 Hi
2 Hi
1 Hi
```

```
#Print all the even numbers between 0 to 10 (both inclusive) using a while loop

i = 0
while i<= 10:
  print(i)
  i += 2
```

```
0
2
4
6
8
10
```

```
i = 0
while i<= 10:
  if i%2 == 0:
    print(i)
  i += 1
```

```
0
2
4
6
8
10
```

```
#Print all the odd numbers between 0 to 10.

i = 1
while i<= 9:
  print(i)
  i += 2
```

```
1
3
5
7
9
```

```
i = 0
while i<= 10:

  if i%2 != 0:
    print(i)

  i += 1
```

```
1
3
5
7
9
```

```
#Print the sum of all the numbers between 1 to 10 (both inclusive)
# 1. for loop
# 2. while loop
```

```
sum = 0
for i in range(1,11):
  sum += i
  print(sum)
```

```
1
3
```

```
6
10
15
21
28
36
45
55
```

```
# n = 10, 10*11/2 = 55
```

```
sum = 0
i = 1

while i<= 10:
  sum += i
  i += 1

print(sum)
```

```
55
```

```
#Print the multiplication table of 5 till its first 10 multiples:

5*1 = 5
5*2 = 10
...
...
```

```
n = 5
i = 1

while i<= 10:
  print(n,"*",i, "=", n*i)
  i += 1
```

```
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

Quiz + Practice

```
What is the output?
count = 1

while count <= 3:
    print(count)
    count += 1

A.

1
2
3

B.

1
2
3
4

C.

0
```

```
1
2

D. Infinite Loop

Ans: A
```

```
How many times will the following loop execute?
count = 5

while count <= 10:
    print(count)
    count += 1

A. 5

B. 6

C. 10

D. 12
```

```
count = 5

while count <= 10:
    print(count)
    count += 1
```

```
5
6
7
8
9
10
```

```
What is the output?
count = 10

while count >= 8:
    print(count)
    count -= 1
```

```
count = 10

while count >= 8:
    print(count)
    count -= 1
```

```
10
9
8
```

```
Which statement is TRUE about a while loop?

A. It always runs exactly 10 times.

B. It runs as long as its condition is True.

C. It can only print numbers.

D. It always requires range().

Ans: B
```

```
What is the output?
count = 1

while count < 1:
    print("Hello")

A.

Hello
```

B.

Hello
Hello

C. Nothing

D. Error

Ans: C

```
count = 1

while count < 1:
    print("Hello")
```

Which of the following is the correct syntax?

A.

```
while i < 5
```

B.

```
while(i < 5)
```

C.

```
while i < 5:
```

D.

```
while i = 5:
```

Ans: C

What will this code print?

```
num = 2

while num <= 8:
    print(num)
    num += 2
```

A.

2
4
6
8

B.

2
4
6
8
10

C.

1
3
5
7

D. Infinite Loop

Ans: A

What is the output?

```
count = 5
```

```
while count > 0:
    print(count)
    count -= 2

A.

5
3
1

B.

5
4
3
2
1

C.

5
3
1
-1

D. Infinite Loop


Ans: A
```

HW

```
What is the output?
count = 1

while count <= 5:
    count += 1

print(count)

A. 5

B. 6

C. 7

D. Error
```

```
Which situation is best suited for a while loop?

A. Print numbers from 1 to 10.

B. Print multiplication table of 5.

C. Keep asking for a password until the correct one is entered.

D. Print the first 20 even numbers.
```

```
What is the output?
x = 0

while x < 3:
    print(x)
    x += 1

print("Done")

A.

0
1
2
Done
```

```
B.


1
2
3
Done

C.

0
1
2
3
Done
```